

VLINK AI HACKATHON

IncidentIQ

Autonomous Incident Command Center Powered by Multi-Agent AI

An automated command center that reduces Mean Time to Resolution (MTTR) to under 90 seconds — coordinating AI agents to triage, diagnose, mitigate, and write post-mortems for outages.

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Executive Summary

The Next Generation of Incident Operations

IncidentIQ is an autonomous operational system that intercepts system alerts, correlates telemetry, and resolves issues — **eliminating human delays** in the operational loop.



Core Capabilities



6 Specialized AI Agents

Sequential crew running on Gemini 1.5 Pro, executing complex diagnostic logic.



Real-Time Telemetry

Integrates live metrics, historical outage DBs, and Confluence runbooks.



Zero Human Handoff

Full lifecycle automation from initial alert trigger to final post-mortem.

The SRE Challenge

The Broken Incident Response Pipeline

Alert Fatigue

Operations engineers are flooded with hundreds of uncorrelated alerts daily, masking critical incidents.

High MTTR

Finding root cause, searching wikis, and applying fixes manually takes hours every single time.

Knowledge Silos

Prior incident logs and engineer comments are buried and rarely shared across teams.

Missing Documentation

Writing blameless post-mortems is often delayed, causing organizations to lose critical learning.



The Solution: IncidentIQ

Fully Autonomous Incident Lifecycle Control

IncidentIQ intercepts every alert, runs a coordinated AI crew through diagnosis and remediation, and closes the loop with auto-generated documentation — **all in under 90 seconds.**



1. Intercept

Alert Ingestion



2. Diagnose

Multi-Agent RCA



3. Mitigate

Parametrized Fix



4. Close

Zero-Touch Comms

Enterprise Technology Stack

Built for Speed, Typesafety, and Reliability



Core Backend

Python 3.11 + FastAPI on Uvicorn for high-performance async operations.



SRE Agent Framework

CrewAI + Gemini 1.5 Pro for sequential, persona-driven agent workflows.



Vector Database

ChromaDB storing vectorized Confluence runbooks for semantic RAG retrieval.



Frontend Interface

React.js + Tailwind CSS delivering glassmorphic dark-theme SRE dashboards.

SRE Multi-Agent Crew

6 Persona-Based AI Agents Collaborating Sequentially

01. Triage Agent

Classifies alert severity and verifies SLA windows.

02. RCA Agent

Pulls metrics and compares telemetry against past outages.

03. Runbook Agent

Semantic vector search on wikis to find the correct recovery guide.

04. Comms Agent

Moderates Slack channels and broadcasts status cards.

05. Jira Agent

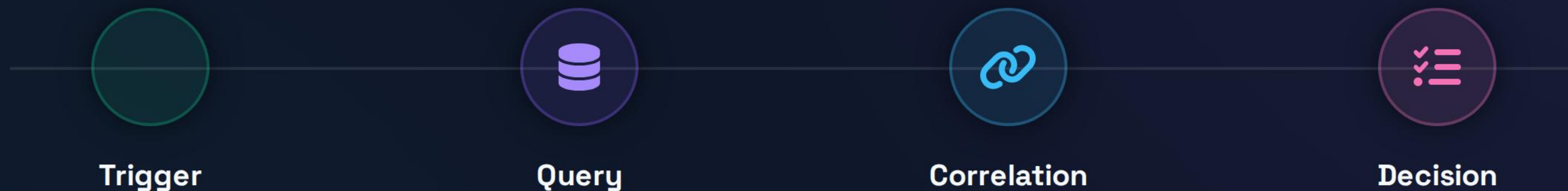
Creates tickets and records deployment change logs.

06. Post-Mortem Agent

Automatically generates structured blameless post-mortems.

Diagnostic Workflow & Pipeline

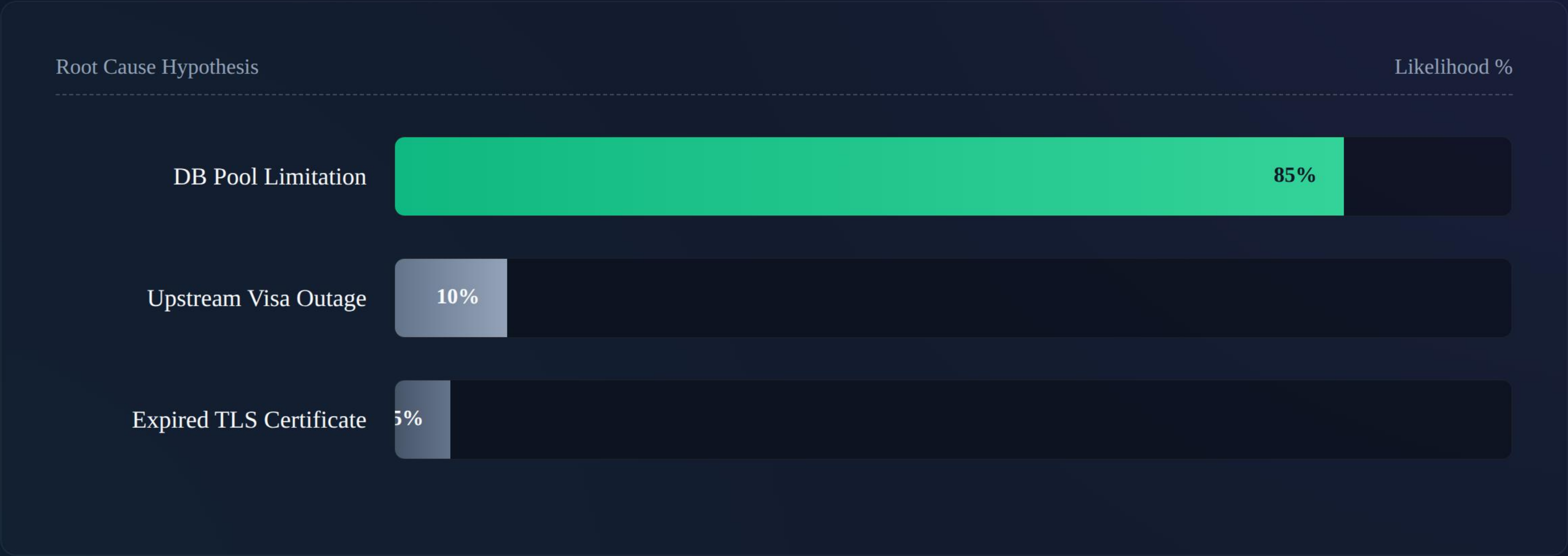
Step-by-Step Reasoning Flow



The **RCA Agent** drives comparative analysis, outputting likelihood weightings to eliminate operational guesswork.

Comparative RCA Output

Data-Driven Root Cause Identification



IncidentIQ outputs weighted likelihood scores, clearly rules out alternative causes, and recommends secondary fallbacks if primary mitigations fail.

Runbook Adaptation & Sync

Semantic RAG Retrieval + Enterprise Tool Sync



1. ChromaDB Playbook Index

Confluence playbooks vectorized for semantic similarity matching.



2. Context Adaptation

Generic wiki instructions translated into parameter-mapped commands with live IDs.



3. Toolchain Integration

Auto-creates Slack channels, logs Jira tickets with SLA bounds, writes post-mortems.



4. Real-Time Dashboard

Glassmorphic WebSocket console broadcasting agent steps and live preview panels.



Live Operations Console

Real-time agent observation and metric correlation.

Business ROI & SLA Gains

Maximizing Operations Efficiency

2s

MTTD

Mean Time to Detection — down
from 15 minutes

90s

MTTR

Mean Time to Resolution —
down from 90 minutes

0_{min}

Doc Overhead

Post-mortem writing time —
down from hours of manual
effort

✓ Team burnout significantly minimized by automatically filtering noisy alerts and alert storms.

Stability & Roadmap

Scaling IncidentIQ Enterprise-Wide

Demo Reliability Strategy

Gemini API free tier limits requests to 15 RPM. IncidentIQ pre-seeds a local JSON resolution cache mapping prompt hashes to responses — enabling 100% offline operation with zero failures during live pitches.

Future Roadmap

Live Integrations:

Direct hooks to Kubernetes clusters and Terraform state.

Self-Healing Actions:

Automated canary rollbacks and dynamic replica scaling.

 Abstract futuristic glowing nodes network diagram



Questions?

Thank you for your time and attention.

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Image Sources



<https://ram-tech.co.il/wp-content/uploads/2023/01/VLINK.png>

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